

Md. Wahidur Rahman

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🔍 Google Scholar

🌐 LinkedIn

🌐 Portfolio

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Education

- 2025 – Present 📖 **Ph.D. Engineering, Texas A&M University - Kingsville** Major in Computer Science.
Status: *Ongoing*;
Topic: *Cybersecurity in Health Informatics*
- 2021 – 2024 📖 **M.Sc. Computer Science and Engineering, Mawlana Bhashani Science and Technology University (MBSTU)** in Computer Science and Engineering (CSE).
CGPA: *4.00 (Out of 4.00)*; (Last two years CGPA: *3.86*)
Place: *1st* Position
- 2015 – 2019 📖 **B.Sc. Computer Science and Engineering, Mawlana Bhashani Science and Technology University (MBSTU)** in Computer Science and Engineering (CSE).
CGPA: *3.78 (Out of 4.00)*
Place: *4th* Position

Employment History

- 2025 – Now 📖 **Graduate Assitant Research**, Department of Electrical Engineering & Computer Science, Texas A&M University-Kingsville, Texas 78363.
- 2023 – Now 📖 **Founder & CEO**, Wahid's Research Lab, Dhaka, Bangladesh.
Website: <https://wreslab.com/>
- 2024 – On Leave 📖 **Assistant Professor**, Department of Computer Science and Engineering, Uttara University, Dhaka, Bangladesh.
- 2022 – 2024 📖 **Senior Lecturer & Coordinator**, Department of Computer Science and Engineering, Uttara University, Dhaka, Bangladesh.
- 2022 – 2023 📖 **Lecturer**, Department of Computer Science and Engineering, Uttara University, Dhaka, Bangladesh.
- 2022 – 2025 📖 **Adjunct Lecturer**, Department of Computer Science and Engineering, Bangladesh University, Dhaka, Bangladesh.
- 2021 – 2022 📖 **Lecturer**, Department of Computer Science and Engineering, Khwaja Yunus Ali University, Sirajganj, Bangladesh.
- 2020 – 2022 📖 **Senior Researcher**, Time Research & Innovation (TRI), Dhaka, Bangladesh.
Website: <https://timerni.com/>

Research Interest

- 📖 **Computer Vision and Medical Imaging**
- 📖 **IoT and Deep Learning in Agriculture**
- 📖 **Machine Learning and Deep Learning**
- 📖 **Data Privacy and Security in AI**

Research Publications

Journal Articles

- 1 Rajbongshi, Aditya, Fatema Tuz Johora, Arafat Hossain, Md Salauddin Sarker, Md Habibur Rahman, Md Wahidur Rahman, Fahad T Alotaibi, and Mohammad Ali Moni. "Leveraging explainable AI for sustainable agriculture: a comprehensive review of recent advances". *Artificial Intelligence Review* (2026).

- 2 Al Reshan, Mana Saleh, Wahidur Rahman, Shisir Mia, Mehedi Hasan Talukder, Mohammad Motiur Rahman, Asadullah Shaikh, Tunc Asuroglu, and Jawad Rasheed. "An AI-powered smart Agribot for detecting locusts in farmlands using IoT and deep learning". *Scientific Reports* 15, no. 1 (2025): 39848.
- 3 Alam, Elman, Md Tarequl Islam, Ishrat Zahan Raka, Onamika Sarkar Ritu, Md Shakhawat Hossain, **Wahidur Rahman**, and Rahat Khan. "Ripen banana dataset: A comprehensive resource for carbide detection and ripening stage analysis to enhance food quality and agricultural efficiency". *Data in Brief* 60 (2025): 111659. ISSN: 2352-3409. <https://doi.org/10.1016/j.dib.2025.111659>.
- 4 Kaysar, Md Nadim, Mahjabin Namira Mazumder, Wahidur Rahman, Al Shahriar Rubel, and Muhammad Suleman. "Enhanced Water Quality Index Prediction through Machine Learning and IoT with Hybrid Voting Methods". *Water Resources* 52, no. 6 (2025): 1340–1355.
- 5 Mim, Mithila Akter, M. R. Khatun, **Wahidur Rahman** Hossain Muhammad Minoar, and Arslan Munir. "Exploring Early Learning Challenges in Children Utilizing Statistical and Explainable Machine Learning". *Algorithms* 18, no. 1 (2025): 20. <https://doi.org/10.3390/a18010020>.
- 6 Mim, Mithila Akter, Mst. Rokeya Khatun, Muhammad Minoar Hossain, and **Wahidur Rahman**. "Advanced PCF-SPR biosensor design and performance optimization using machine learning techniques". *Optik* 333 (2025): 172391. ISSN: 0030-4026. <https://doi.org/10.1016/j.ijleo.2025.172391>.
- 7 Rahman, Md Wahidur, Md Rahat Khan, and Mais Nijim. "Privacy-Preserving Deep Learning for Disease Diagnosis in Medical Imaging: A Systematic Review". *IEEE Access* (2025).
- 8 Shaikh, Asadullah, **Wahidur Rahman**, Kaniz Rokhsana, Tarequl Islam, Mohammad Motiur Rahman, Hani Alshahrani, Adel Sulaiman, and Mana Saleh Al Reshan and. "An improved deep CNN-based freshwater fish classification with cascaded bio-inspired networks". *Automatika* 66, no. 2 (2025): 249–280. <https://doi.org/10.1080/00051144.2025.2457803>. eprint: <https://doi.org/10.1080/00051144.2025.2457803>.
- 9 Tania, Nusrat, Md Mijanur Rahman, AHM Saifullah Sadi, and Wahidur Rahman. "Pain Intensity Recognition from Facial Expression Using Deep Learning". *Statistics, Optimization & Information Computing* 14, no. 5 (2025): 2590–2604.
- 10 Bappy, Md Tusher Ahmad, Kazi Mehedi Hasan Rabbi, **Wahidur Rahman** Ahmed Md Jonayed, Mahin Zeesan, AHM Saifullah Sadi, and Mohammad Motiur Rahman. "SeasVeg: An image dataset of Bangladeshi seasonal vegetables". *Data in Brief* 55 (2024): 110564. <https://doi.org/10.1016/j.dib.2024.110564>.
- 11 Elmagzoub, M. A. **Wahidur Rahman**, Kaniz Rokhsana, Md Tarequl Islam, AHM Saifullah Sadi, Mohammad Motiur Rahman, Adel Rajab, Khairan Rajab, and Asadullah Shaikh. "Paddy Insect Identification using Deep Features with Lion Optimization Algorithm". *Heliyon* (2024). <https://doi.org/10.1016/j.heliyon.2024.e32400>.
- 12 Hai, Talha Bin Abdul, Wahidur Rahman, Md Solaiman Hosen, Md Tarequl Islam, AHM Saifullah Sadi, Gazi Golam Faruque, and Mir Mohammad Azad. "Smart Security Solution for Market Shop Using IoT and Deep Learning". *Indonesian Journal of Electrical Engineering and Informatics (IJEI)* 12, no. 1 (2024): 195–205. <https://doi.org/10.52549/ijeai.v12i1.4780>.
- 13 Hossain, Muhammad Minoar, Md Abul Ala Walid, SM Saklain Galib, Mir Mohammad Azad, Wahidur Rahman, A. S. M. Shafi, and Mohammad Motiur Rahman. "COVID-19 detection from chest CT images using optimized deep features and ensemble classification". *Systems and Soft Computing* (2024): 200077. <https://doi.org/10.1016/j.sasc.2024.200077>.
- 14 Islam, Md Tarequl, Wahidur Rahman, Md Shakhawat Hossain, Kaniz Rokhsana, Irma Domínguez Azpíroz, Raquel Martínez Diaz, Imran Ashraf, and Md. Abdus Samad. "Medicinal Plant Classification Using Particle Swarm Optimized Cascaded Network". *IEEE Access* (2024). <https://doi.org/10.1109/access.2024.3378262>.

- 15 Jahin, Iftasam Islam, Munni Khatun, Md Tarekul Islam, Md Wahidur Rahman, and Ishrat Zahan Raka. "BDHusk: A comprehensive dataset of different husk species images as a component of cattle feed from different regions of Bangladesh". *Data in Brief* 52 (2024): 110018. <https://doi.org/10.1016/j.dib.2023.110018>.
- 16 Rahman, Wahidur, Mohona Akter, Nahida Sultana, Maisha Farjana, Arfan Uddin, Md Bakhtiar Mazrur, and Mohammad Motiur Rahman. "BDWaste: A comprehensive image dataset of digestible and indigestible waste in Bangladesh". *Data in Brief* (2024): 110153. <https://doi.org/10.1016/j.dib.2024.110153>.
- 17 Rahman, Wahidur, Muhammad Minoar Hossain, Md Mahedi Hasan, Md Sadiq Iqbal, Mohammad Motiur Rahman, Khondokar Fida Hasan, and Mohammad Ali Moni. "Automated Detection of Harmful Insects in Agriculture: A Smart Framework Leveraging IoT, Machine Learning, and Blockchain". *IEEE Transactions on Artificial Intelligence* (2024). <https://doi.org/10.1109/TAI.2024.3394799>.
- 18 **Wahidur Rahman**, Mohammad Motiur Rahman, Md Ariful Islam Mozumder, Rashadul Islam Sumon, Samia Allaoua Chelloug, Rana Othman Alnashwan, and Mohammed Saleh Ali Muthanna. "A Deep CNN-Based Salinity and Freshwater Fish Identification and Classification Using Deep Learning and Machine Learning". *Sustainability* 16, no. 18 (2024): 7933. <https://doi.org/10.3390/su16187933>.
- 19 Ahmed, Md Asif, Md Shakil Hossain, Wahidur Rahman, Abdul Hasib Uddin, and Md Tarekul Islam. "An advanced Bangladeshi local fish classification system based on the combination of deep learning and the internet of things (IoT)". *Journal of Agriculture and Food Research* (2023): 100663. <https://doi.org/10.1016/j.jafr.2023.100663>.
- 20 Alom, Munjur, Md Yeasin Ali, Md Tarekul Islam, Abdul Hasib Uddin, and Wahidur Rahman. "Species classification of brassica napus based on flowers, leaves, and packets using deep neural networks". *Journal of Agriculture and Food Research* 14 (2023): 100658. <https://doi.org/10.1016/j.array.2023.100292>.
- 21 Elias Hossain, **Wahidur Rahman**, Abdullah Alshahrani. "News Modeling and Retrieving Information: Data-Driven Approach". *Intelligent Automation & Soft Computing* 38, no. 2 (2023): 109–123. ISSN: 2326-005X. <https://doi.org/10.32604/iasc.2022.029511>.
- 22 Kabir, Hasnain, Taslima Juthi, Md Tarekul Islam, Md Wahidur Rahman, and Rahat Khan. "WaterHyacinth: A Comprehensive Image Dataset of Various Water Hyacinth Species From Different Regions of Bangladesh". *Data in Brief* (2023): 109872. <https://doi.org/10.1016/j.dib.2023.109872>.
- 23 Rahman, Wahidur, Mohammad Gazi Golam Faruque, Kaniz Roksana, AHM Saifullah Sadi, Mohammad Motiur Rahman, and Mir Mohammad Azad. "Multiclass blood cancer classification using deep CNN with optimized features". *Array* 18 (2023): 100292. <https://doi.org/10.1016/j.array.2023.100292>.
- 24 Hossain, Elias, A. A. Almakdi, Mohammed Alshehri, M. Gazi Golam Faruque, and Wahidur Rahman. "Forecasting Mental Stress Using Machine Learning Algorithms". *Computers, Materials & Continua* (2022). <https://doi.org/10.32604/cmc.2022.027812>.
- 25 Hossain, Elias, M. A. Almakdi, Hanan Halawani, Md. Mizanur Rahman, Wahidur Rahman, Sabila Al Jannat, Nadim Kaysar, and Shishir Mia. "Intelligent Health Monitoring Tool: Diabetes Diagnosis using Machine Learning Analysis with Smartphone Application". *Computers, Materials & Continua* (2022). <https://doi.org/10.32604/cmc.2022.027115>.
- 26 Hossain, Muhammad Minoar, Reshma Ahmed Swarna, Rafid Mostafiz, Pabon Shaha, Lubna Yasmin Pinky, Mohammad Motiur Rahman, Wahidur Rahman, Md. Selim Hossain, Md. Elias Hossain, and Md. Sadiq Iqbal. "Analysis of the performance of feature optimization techniques for the diagnosis of machine learning-based chronic kidney disease". *Machine Learning with Applications* 9 (2022): 100330. <https://doi.org/10.1016/j.mlwa.2022.100330>.
- 27 Rahman, Hasibur, Md. Omar Faruq, Talha Bin Abdul Hai, Wahidur Rahman, Muhammad Minoar Hossain, Mahbulul Hasan, Shafiqul Islam, Md. Moinuddin, Md. Tarekul Islam, and Mir Mohammad Azad. "IoT enabled mushroom farm automation with Machine Learning to classify toxic mushrooms in Bangladesh".

Journal of Agriculture and Food Research 7 (2022): 100267. <https://doi.org/10.1016/j.jafr.2021.100267>.

- 28 Rahman, Md. Wahidur, Rahabul Islam, Arafat Hasan, Nasima Islam Bithi, Md. Mahmodul Hasan, and Mohammad Motiur Rahman. "Intelligent waste management system using deep learning with IoT". *Journal of King Saud University - Computer and Information Sciences* 34, no. 5 (2022): 2072–2087. <https://doi.org/10.1016/j.jksuci.2020.08.016>.
- 29 Rahman, Wahidur, Abdulwahab Alazeb, Muhammad Minoar Hossain, Saima Siddique Tashfia, Md. Tarekul Islam, Shisir Mia, and Mohammad Motiur Rahman. "IoT and Blockchain-based Mask Surveillance System for COVID-19 Prevention Using Deep Learning". *Computers, Materials & Continua* (2022). <https://doi.org/10.32604/cmc.2022.027126>.
- 30 Shah Siddiqui, Murshedul Arifeen, Adrian Hopgood, Alice Good, Alexander Gegov, Elias Hossain, Wahidur Rahman, Shazzad Hossain, Sabila Al Jannat, Rezowan Ferdous, and Shamsul Masum. "Deep Learning Models for the Diagnosis and Screening of COVID-19: A Systematic Review". *SN Computer Science* 3, no. 5 (2022): 397. <https://doi.org/10.1007/s42979-022-01326-3>.
- 31 Islam, Rahabul, Md. Wahidur Rahman, Rahmina Rubaiat, Md. Mahmodul Hasan, Md. Mahfuz Reza, and Mohammad Motiur Rahman. "LoRa and server-based home automation using the internet of things (IoT)". *Journal of King Saud University - Computer and Information Sciences* (2021). <https://doi.org/10.1016/j.jksuci.2020.12.020>.
- 32 Rahman, Md. Wahidur, Saima Siddique Tashfia, Rahabul Islam, Md. Mahmodul Hasan, Sadee Ibn Sultan, Shisir Mia, and Mohammad Motiur Rahman. "The architectural design of smart blind assistant using IoT with deep learning paradigm". *Internet of Things* 13 (2021): 100344. <https://doi.org/10.1016/j.iot.2020.100344>.
- 33 Rahman, Md. Wahidur, Md. Elias Hossain, Rahabul Islam, Md. Harun Ar Rashid, Md. Nur A Alam, and Md. Mahmodul Hasan. "Real-time and low-cost IoT based farming using raspberry Pi". *Indonesian Journal of Electrical Engineering and Computer Science* 17, no. 1 (2020): 197–204. <https://doi.org/10.11591/ijeecs.v17.i1.pp197-204>.
- 34 Rahman, Md. Wahidur, Rahabul Islam, Md. Mahmodul Hasan, Shisir Mia, and Mohammad Motiur Rahman. "IoT Based Smart Assistant for Blind Person and Smart Home Using the Bengali Language". *SN Computer Science* 1 (2020): 300. <https://doi.org/10.1007/s42979-020-00317-6>.
- 35 Hossain, Md. Elias, Md. Wahidur Rahman, Md. Tarekul Islam, and Md. Selim Hossain. "Manifesting a mobile application on safety which ascertains women salus in Bangladesh". *International Journal of Electrical and Computer Engineering* 9, no. 5 (2019): 4355–4363. <https://doi.org/10.11591/ijece.v9i5.pp4355-4363>.

Conference Proceedings

- 1 Mohd, Abdul Kareem Uddin, Md Habibur Rahman, Md Wahidur Rahman, Avdesh Mishra, Tarek Mahmud, and Maleq Khan. "GXMalDetect: A Hybrid GA-XGBoost Architecture for Malware Detection Using Static Image Features". In *2026 IEEE 5th International Conference on AI in Cybersecurity (ICAIC)*, 1–6. IEEE, 2026.
- 2 Rahman, Md Habibur, Md Wahidur Rahman, Avdesh Mishra, Tarek Mahmud, and Mais Nijim. "TabNet-IDS: A TabNet-Driven Tabular Deep Learning Framework for Intrusion Detection Systems". In *2026 IEEE 5th International Conference on AI in Cybersecurity (ICAIC)*, 1–6. IEEE, 2026.
- 3 Rahman, Md Wahidur, Mehdi Hasan, Mais Nijim, Muhammad Armughan Ul Haq, and Fawaz Ali Mohammed. "Privacy-Preserving Federated Deep Learning for Nomophobia Risk Prediction From Smartphone Usage Logs". In *2026 IEEE 5th International Conference on AI in Cybersecurity (ICAIC)*, 1–6. IEEE, 2026.

- 4 Rahman, Md Wahidur, Md Habibur Rahman, Avdesh Mishra, Mais Nijim, Ayush Goyal, and David Hicks. "ASCTM: A Swarm Optimization-Enhanced LSTM Framework for Intelligent Intrusion Detection". In *2025 Cyber Awareness and Research Symposium (CARS)*, 1–6. IEEE, 2025.
- 5 Islam, Md Sariful, Robaaed Afred Tanik, Noor Hossain, Md Abdur Rahman Tamim, Mohammed Imran, Jubayer Ahmed Shawon, Wahidur Rahman, and Samia Yasmin. "Automated Grading of Bananas Using Advanced YOLO Models: A Machine Learning and Computer Vision Approach for Ripeness Classification". In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 3134–3139. IEEE, 2024.
- 6 Islam, Md. Sariful, Md Tusher Ahmad Bappy, Jubayer Ahmed Shawon, Mehedi Hasan, Wahidur Rahman, Lija Akter, and Mir Mohammad Azad. "Cat-CNN: Human Eye Cataract Detection from Color Fundus Photograph with Deep CNN with Optimized Cascaded Network". In *2024 IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, 1–6. 2024. <https://doi.org/10.1109/SPICSCON64195.2024.10941258>.
- 7 Rahman, Wahidur, Md Tusher Ahmad Bappy, Md Sariful Islam, Jubayer Ahmed Shawon, Kamrunnahar Mim, Mohammad Kaiser Ishaque, Md Imran Hossain, and Razib Kumer Debnath. "Cascaded Deep CNN Pipeline for Pneumonia Screening: An Explainable AI based Application". In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2056–2061. IEEE, 2024.
- 8 Rahman, Wahidur, Md Shadman Sakib, Md Solaiman Hosen, Mir Mohammad Azad, Hani Alshahrani, Adel Sulaiman, Mana Saleh Al Reshan, and Asadullah Shaikh. "Cascaded Deep CNN Network for Breast Cancer Screening from Ultrasound Images Using Differential Evolution Algorithm". In *2024 4th International Conference on Digital Society and Intelligent Systems (DSInS)*, 194–199. 2024. <https://doi.org/10.1109/DSInS64146.2024.10992070>.
- 9 Shuvo, Eshat Ahmad, Wahidur Rahman, Md Solaiman Hosen, Mir Mohammad Azad, Mana Saleh Al Reshan, Adel Sulaiman, Hani Alshahrani, and Safeeullah Soomro. "Optimized Hybrid Cascaded Approach for Accurate OSCC Detection in Histopathology Images Using Deep CNNs". In *International Conference on Innovation of Emerging Information and Communication Technology*, 111–124. Springer, 2024.
- 10 Shuvo, Eshat Ahmad, **Wahidur Rahman**, Md Shakhawat Hossain, Md Tarequl Islam, and Md Sadiq Iqbal. "Bio-inspired Heuristic Optimization-based Cascaded Network for Diabetic Retinopathy Screening". In *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, 1–6. IEEE, 2024. <https://doi.org/10.1109/ICAEEE62219.2024.10561821>.
- 11 **Wahidur Rahman**, Riasaad Haque Aneek, Md Moinuddin, Md Shadman Sakib, Md Sadiq Iqbal, and Mohammad Motiur Rahman. "Cardiovascular Disease Prediction Utilizing Machine Learning and Feature Selection with Clonal Selection Algorithm". In *2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, 1–6. IEEE, 2023. <https://doi.org/10.1109/STI59863.2023.10464949>.
- 12 **Wahidur Rahman**, Md Abul Ala Walid, SM Saklain Galib, Kaniz Rokhsana, Talha Bin Abdul Hai, Mir Mohammad Azad, and Mohammad Ali Moni. "Observation of Heart Attack Patients Utilizing Machine Learning with Monarch Butterfly Optimization and IoT". In *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 1–6. IEEE, 2023. <https://doi.org/10.1109/ICCIT60459.2023.10441444>.
- 13 Kamruzzaman, M. M., Md. Moinuddin, Atikul Islam Liton, Mir Mohammad Azad, Muhammad Anwar Hossain, and **Wahidur Rahman**. "Evaluating the Performance of State-of-the-art Methods and Classifying Covid-19 Infected Tissues". In *2022 IEEE 7th International conference for Convergence in Technology (I2CT)*, 1–6. IEEE, 2022. <https://doi.org/10.1109/I2CT54291.2022.9824742>.

Books and Chapters

- 1 Islam, Md. Tarequl, Md. Selim Azad, Md. Sobuj Ahammed, **Md. Wahidur Rahman**, Mir Mohammad Azad, and Mostofa Kamal Nasir. "IoT Enabled Virtual Home Assistant Using Raspberry Pi". In *Distributed*

Computing and Optimization Techniques, vol. 903. Lecture Notes in Electrical Engineering. Springer, Singapore, 2022.  https://doi.org/10.1007/978-981-19-1012-8_55.

- 2 Siddiqui, Shah, Elias Hossain, S. M. Asaduzzaman, Sabila Al Jannat, Ta-seen Niloy, Wahidur Rahman, Shamsul Masum, Adrian Hopgood, Alice Good, and Alexander Gegov. "Analysing and Identifying COVID-19 Risk Factors Using Machine Learning Algorithm with Smartphone Application". In *Inventive Systems and Control*, 775–788. Springer, Singapore, 2022.  https://doi.org/10.1007/978-981-19-1012-8_55.
- 3 Siddiqui, Shah, Elias Hossain, Rezowan Ferdous, Murshedul Arifeen, Wahidur Rahman, Shamsul Masum, Adrian Hopgood, Alice Good, and Alexander Gegov. "A Comparative Study of Deep Learning Models for COVID-19 Diagnosis Based on X-Ray Images". In *Smart and Sustainable Technology for Resilient Cities and Communities*, 163–174. Springer, Singapore, 2022.  https://doi.org/10.1007/978-981-16-9101-0_12.

Research Profile

- 📖 **Google Scholar:** <https://scholar.google.com/citations?user=lvpQZsMAAAAJ&hl=en>
- 📖 **ResearchGate:** <https://www.researchgate.net/profile/Wahidur-Rahman-12>
- 📖 **ORCID:** <https://orcid.org/0000-0001-6115-2364>
- 📖 **Scopus Author ID:** 57202040801

Research Fellowship

- M.Sc. Research 📖 **Thesis Title:** A Deep CNN-based Salinity and Freshwater Fish Identification and Classification using Deep Learning and Machine Learning.
Description: Developed a robust convolutional neural network (CNN) model to accurately identify and classify various species of salinity and freshwater fish. The model leverages deep feature extraction techniques and machine learning classifiers to improve aquatic biodiversity monitoring and support sustainable fishery management. The research addresses challenges such as inter-species similarity and environmental variability.
Associated Publication: 01
Journal Name and Status: *Sustainability*, MDPI [Published] 
Amount: Approximately \$500
Awarded by: National Science and Technology Fellowship, January 2023
Supervised by: Professor Dr. Mohammad Motiur Rahman
Research Location: Mawlana Bhashani Science and Technology University, Bangladesh.
- B.Sc. Research 📖 **Thesis Title:** The Development of Smart System using IoT and Deep Learning Paradigm to detect insects in the agricultural field.
Description: Designed and implemented an IoT-enabled smart monitoring system integrated with deep learning algorithms to detect and classify harmful insects in crop fields. The system utilizes sensor networks and real-time data processing to enable early pest detection, thus helping farmers reduce pesticide use and improve crop yields. The project emphasizes scalability and cost-effectiveness for practical agricultural deployment.
Associated Publication: 02
Journal Name and Status: *IEEE Transactions on Artificial Intelligence & Heliyon* [Published] 
Amount: Approximately \$2500
Awarded by: University Grant Commission (UGC), Bangladesh, January 2020
Supervised by: Professor Dr. Mohammad Motiur Rahman

Technical Skills

Languages	📌 C, C++, C#, Java, Matlab, Python, and R.
ML & DL	📌 Pandas, Numpy, Sk-Learn, Matplotlib, PyTorch, Keras, Tensorflow.
Web Programming	📌 HTML, CSS, JavaScript, Wordpress, Flask, and Django.
IoT Programming	📌 Arduino, Raspberry Pi, LoRa, ESP NodeMCU.
Database	📌 SQL, MySQL, PostgreSQL, Oracle, MariaDB, MongoDB, Redis.
Others	📌 Latex, UML, MS Office, Audacity, Filmora, etc.

Professional Certificates and Membership

Certificate	📌 Skill Development for Mobile Game and Application Project Conducted by the ICT Ministry of Bangladesh May 2018 – November 2018 <i>With Certificate</i>
Membership	📌 IEEE Computer Society , Uttara University Chapter, Bangladesh Section June 2024 – Present Member ID: 96274430

Awards and Achievements

December, 2017	📌 Champion in Project Showcasing , CSE Carnival, Varendra University, Bangladesh. Presented an IoT-based smart farming solution that stood out among 30+ teams for its innovation and real-world applicability.
January, 2018	📌 1st Runners Up in Project Showcasing , EEE Day, Bangladesh Army International University of Science and Technology, Bangladesh. Recognized for a low-cost insect detection system using deep learning and microcontroller integration.
February, 2019	📌 1st Runners Up in Project Showcasing , CSE Carnival, Mawlana Bhashani Science and Technology University, Bangladesh. Demonstrated an AI-powered agricultural drone concept for crop monitoring and analysis.
December, 2019	📌 1st Runners Up in Project Showcasing , CSE TechFest, Varendra University, Bangladesh. Highlighted a portable smart system for salinity-based fish classification using CNN models.
June, 2023	📌 Best Researcher in Computer Science (2022) , Uttara University, Uttara, Dhaka. Awarded for impactful research contributions in AI and IoT-based agriculture and medical diagnosis.
May, 2024	📌 Best Researcher in Computer Science (2023) , Uttara University, Uttara, Dhaka. Recognized for high research output and publications in top-tier journals in computer vision and health informatics.
April, 2026	📌 2nd Position in Poster Presentation , Texas A&M University-Kingsville, USA. Texas A&M University–Kingsville hosted Research and Innovation Day 2026 on April 22 at the Memorial Student Union Building, featuring student poster presentations, collaboration, and research awards..

English Proficiency

- **IELTS Proficiency Level:** B2
IELTS Overall Band Score: 6.00; (*Reading: 6.0, Listening: 6.0, Speaking: 6.0, Writing: 6.0*)

Extra-Curricular Activities

- May 2018 –
December 2021
- **Convener, BASIS Student Forum (BSF)**, Mawlana Bhashani Science and Technology University (MBSTU), Bangladesh.
Led the university chapter organizing tech events, workshops, and student ICT engagement activities.
Website: <https://bsf.basis.org.bd/university/Mawlana%20Bhashani%20Science%20&%20Technology%20University>

References

- **Dr. Mais Nijim**
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- **Dr. Mohammad Motiur Rahman**
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